

Na Min An

PH.D. STUDENT

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Research Interests

My research aims to explore: *how do omni-modal AI systems perceive, integrate, and reason over the full spectrum of human sensory experience?* I am deeply interested in uncovering the mechanisms behind multisensory processing, such as complex textual, audio, and visual understanding, reasoning, and generation.

Education

Korea Advanced Institute of Science and Technology (KAIST)

DOCTOR OF PHILOSOPHY IN ARTIFICIAL INTELLIGENCE, GPA: 4.13/4.3

- Advisors: Prof. Hyunjung Shim (2024-Current) and Dr. James Thorne (2023-2025)

Seoul, South Korea

Feb. 2023-Current

University of Seoul

BACHELOR OF SCIENCE IN MATHEMATICS, GPA: 4.29/4.5

- Summa Cum Laude (1/30) and Received academic excellence or Mayor's scholarship for every semester

Seoul, South Korea

Feb. 2022

Skills

Language & Framework	Python (PyTorch, HuggingFace Transformers), C++, JavaScript, MATLAB, R; Linux, Cloud/Server Computing
NLP & Multimodal LLM	LLM/VLM training, Image Captioning, Text-to-Image Generation, Video Understanding, Multimodal Reasoning
Computer Vision	Scene Understanding, Language-Guided Visual Grounding, Multimodal Retrieval, Data Curation at Scale
Interpretability & Fairness	Mechanistic Interpretability, Human Alignment Metrics
Certifications	Society of Actuaries (P, FM, and SRM)

Experience

KAIST

RESEARCH ASSISTANT

- Ministry of Science and ICT Project: Contributed leadership to a national project aimed at developing socially unbiased generative models.
- Basic Research Lab Project: Co-led a national project on developing barrier-free AI systems in educational and mobility domains.
- Published eight first-author and three co-authored Top-tier works, with research centered on fairness and human-centered ML systems.

Seoul, South Korea

March 2023 - PRESENT

NAVER AI Lab

RESEARCH INTERN

- Analyzing the architecture of omni-modal models and modality effects

Seongnam, South Korea

April 2026 - PRESENT

University of Copenhagen

VISITING RESEARCHER

- Received a Danish Data Science Academy grant to conduct research on interpretability in multimodal models towards fairness-aware systems.
- Developed a neuron-level bias mitigation framework for VLMs and LVLMs, in collaboration with the NVIDIA Taiwan team.

Copenhagen, Denmark

March 2025 - May 2025

Korea Institute of Science and Technology (KIST)

RESEARCH INTERN

- Analyzed human perceptual limits and ML model performance under severely low-resolution visual conditions.
- Developed a reinforcement learning-based framework that emulates the learning effects frequently occurring in human psychophysical tests.
- Built an automated imaging pipeline for detecting fluorescence-image edges and identifying hyperpolarization phenomena with high precision.

Seoul, South Korea

Oct. 2022 - Feb. 2023

University of Seoul

RESEARCH INTERN

- Investigated district-level Covid-19 risk factors across Seoul and awarded 3rd place (out of 40+ teams) in a competitive big-data competition.
- Developed a hybrid CNN-RNN model to classify the progressive stages of pulmonary nodules from sequential CT scan slices.
- Analyzed YouTube channel performance to model subscriber trajectory and provided targeted strategies for subscriber acquisition.

Seoul, South Korea

Sep. 2020 - Dec. 2020

Selected Publications

1. Kim, E.* , **An, N. M.*** , Kang, W. J., Kim, S., Thorne, J., and, Shim, H., “Are Large Vision-Language Models Ready to Guide Blind and Low-Vision Individuals?,” *COLM* (* denotes equal contribution), 2026.
Accessibility Vision-Language Models Evaluation & Benchmarking
2. **An, N. M.*** , Kang, I.* , Lee, M., and Shim, H., “Blind to Position, Biased in Language: Probing Mid-Layer Representational Bias in Vision Language Encoders for Zero-Shot Language-Grounded Spatial Understanding,” *ECCV* (* denotes equal contribution), 2026.
Interpretability Vision-Language Models
3. **An, N. M.**, Jang Y., Hirota, Y., Hachiuma, R., Augenstein, I., and Shim, H., “Interpretable Debiasing of Vision-Language Models for Social Fairness,” *CVPR*, 2026.
Interpretability Fairness & Bias Vision-Language Models
4. Kim, E.* , Park, J.* , **An, N. M.*** , Kim, J., Patel, H. L., Jin, J., Kruk, J., Agarwal, A., Panda, S., Ilasariya, F. A., Shim, H., and Oh, A., “World in a Frame: Understanding Culture Mixing as a New Challenge for Vision-Language Models,” *CVPR* (* denotes equal contribution), 2026.
Fairness & Bias Vision-Language Models Evaluation & Benchmarking
5. Waheed, S.* and **An, N. M.*** , “Image Embedding Sampling Method for Diverse Captioning,” *EMNLP Main* (* denotes equal contribution), 2025.
Multimodal Representation Vision-Language Models
6. **An, N. M.*** , Kim, E.* , Thorne, J., and Shim, H., “IoT: Embedding Standardization Method Towards Zero Modality Gap,” *ACL Outstanding* (Top 1%, 26 out of 3,000 accepted papers, * denotes equal contribution), 2025.
Multimodal Representation Vision-Language Models Evaluation & Benchmarking
7. Bayramli, Z.* , Suleymanzade, A.* , **An, N. M.**, Ahmad, H., Kim, E., Kang, J., Thorne, J., and Oh, A., “Diffusion Models Through a Global Lens: Are They Culturally Inclusive?” *ACL Oral* (Top 8%, 243 out of 3,000 accepted papers, * denotes equal contribution), 2025.
Fairness & Bias Evaluation & Benchmarking
8. Kang, W. J., Kim, E., **An, N. M.**, Kim, S., Choi, H., Kwak, K., and Thorne, J., “Sightation Counts: Leveraging Sighted User Feedback in Building a BLV-aligned Dataset of Diagram Descriptions,” *ACL Main*, 2025.
Accessibility Evaluation & Benchmarking
9. **An, N. M.**, Roh, H., Kim, S., Kim, J. H., and Im, M., “Machine Learning Techniques for Simulating Human Psychophysical Testing of Low-Resolution Phosphene Face Images in Artificial Vision,” *Advanced Science* (Impact Factor: 14.3, JCR Ranking: 6.5%), 2025 - Featured in *Korean National News*.
Accessibility Evaluation & Benchmarking
10. Jung, W., Hong, J., **An, N. M.**, Thorne, J., and Yun, S., “Stable Language Model Pre-training by Reducing Embedding Variability,” *EMNLP Main*, 2024.
Text Representation Language Models
11. Lee, N.* , **An, N. M.*** , and Thorne, J. “Can Large Language Models Infer and Disagree Like Humans?” *EMNLP Main* (* denotes equal contribution), 2023.
Language Models Evaluation & Benchmarking

Publications

1. Park, N.*, **An, N. M.***, Kim, K.*, Yoon, S., Huo, J., and, Shim, H., “Aligned but Stereotypical? The Hidden Influence of System Prompts on Social Bias in LVLM-Based Text-to-Image Models,” *ICLR RSI Workshop* (* denotes equal contribution), 2026.

Fairness & Bias

Vision-Language Models

Evaluation & Benchmarking

2. Waheed, S., **An, N. M.**, Milford, M., Ramchurn, S. D., and Ehsan, S., “VLM-Guided Visual Place Recognition for Planet-Scale Geo-Localization,” *ACRA*, 2025.

Multimodal Representation

Vision-Language Models

3. Kim, J. S.*, Yu, A.*, Ismayilzada, J.*, Park, J., Kim, E., Ahmad, H., **An, N. M.**, Thorne, J., and Oh, A., “WHEN TOM EATS KIMCHI: Evaluating Cultural Awareness of Multimodal Large Language Models in Cultural Mixture Contexts,” *NAACL C3NLP Workshop Outstanding* (* denotes equal contribution), 2025.

Fairness & Bias

Vision-Language Models

Evaluation & Benchmarking

4. **An, N. M.**, Waheed, S., and Thorne, J., “Capturing the Relationship Between Sentence Triplets for LLM and Human-Generated Texts to Enhance Sentence Embeddings,” *EACL Findings*, 2024.

Text Representation

Language Models

5. **An, N. M.**, Roh, H., Kim, S., Kim, J. H., and Im, M., “Reinforcement Learning Framework to Simulate Short-Term Learning Effects of Human Psychophysical Experiments Assessing the Quality of Artificial Vision,” *IJCNN Oral*, 2023.

Accessibility

Evaluation & Benchmarking

6. **An, N. M.**, Roh, H., Jung, S., Kim, E. J., and Im, M., “Machine Learning Approaches as an Alternative to Human Psychophysical Tests of Prosthetic Vision,” *EMBS Abstract*, 2021.

Accessibility

Evaluation & Benchmarking

Patents

1. Im, M., Roh, H., and **An, N. M.**, Kim, J. H., “Artificial Vision Parameter Learning and Automating Method for Improving Visual Prosthetic Systems”, *US Patent Granted No. US 12,602,618 B2*, 2026.
2. Im, M., Roh, H., and **An, N. M.**, Kim, J. H., “Artificial Vision Expression Parameter Automation Learning System Method for Improving Artificial Vision Device”, *Korea Patent Granted No. 0172619*, 2021.
3. **An, N. M.**, Kim, E., Shim, H., “A Fine-Grained Vision-Language Alignment Evaluation Framework Reflecting Users’ Diverse Visual Abilities”, *KAIST Patent Submitted No. P2026-0725*, 2026.

Service

Talks

- AI Tutorial Series at KAIST AI (Winter 2025).

Volunteer

- Provided on-site support for EMNLP conference (2023).
- Contributed to community support and educational outreach in Cambodia in Summer 2018.

Team Leader

- Lab representative student from March 2024 to Aug. 2025.
- Appeared in the KAIST AI promotional video (2024) and top-10 research achievements participant (2023).
- Lab research manager from Aug. 2023 to Dec. 2023.

Teaching Assistant

- Department Seminars & Colloquia, Fall 2023 & Spring 2026
- Computational Image Generation and Modification, Fall 2025

Mentor

- Led research mentorship for two Azerbaijan undergraduate students in South Korea from Sep. 2024 to Jan. 2025.
- Provided English and mathematics tutoring to Dongdaemun-gu (Seoul, South Korea) high school students in Fall 2019 and Winter 2020.

Reviewer

- IEEE Transactions on Audio, Speech and Language Processing 2025 and 2026
- ACL ARR February, May, July, October 2025; January, April, May, July 2026; COLM 2026
- ACL Student Research Workshop 2025 and 2026; ICLR RSI Workshop 2026
- IJCNN 2024, 2025, 2026, and 2027

References

- **Hyunjung Shim**, Associate Professor, KAIST AI | kateshim@kaist.ac.kr
- **Dongyoon Han**, Leader, NAVER AI Lab | dongyoon.han@navercorp.com
- **Taekyung Kim**, Researcher, NAVER AI Lab | taekyungkim918@gmail.com
- **Jaemyung Yu**, Researcher, NAVER AI Lab | jaemyung@kaist.ac.kr
- **Ryo Hachiuma**, Research Scientist, NVIDIA | rhachiuma@nvidia.com
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- **Alice Oh**, Professor, KAIST School of Computing | alice.oh@kaist.edu
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- **Maesoon Im**, Principal Research Scientist, KIST | maesoon.im@kist.re.kr